

Peter Xu

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<https://xretep.github.io/pdfs/>

EDUCATION

2017 - 2023 PhD in Mathematics at **McGill University**

2012 - 2016 BA in Mathematics at the **University of Chicago** (minor: Human Rights)

EMPLOYMENT

2023 - present Hedrick Assistant Adjunct Professor at **UCLA**

SELECTED ARTICLES AND RESEARCH

Symbols for toric Eisenstein cocycles and arithmetic applications

Submitted. Obtains explicit expressions for Eisenstein cocycles for arithmetic group actions on self-products of $\mathbf{G}_m$, by constructing explicit symbols in motivic cohomology complexes. Improves existing Hecke and integrality results for Sharifi's map ϖ_N relating the Eisenstein geometry of $\mathrm{GL}_2(\mathbb{Q})$ and the motivic cohomology of cyclotomic fields. Establishes a duality relationship of the de Rham regulator to smoothed Shintani generating functions, and thereby obtains a new Shintani-style proof of the 2-adic congruences of Deligne–Ribet. Available at <https://xretep.github.io/pdfs/toric.pdf>.

Localization and elliptic motivic relations

Submitted. Derives from a general motivic localization formalism a large generalization of classical Suslin/Weil-style reciprocity laws, including in the setting of motivic homotopy theory. Uses this to construct explicit modular symbols valued in integral-coefficient motivic cohomology of some arithmetic families of group schemes with arbitrary level structures, including symbols valued in Beilinson–Kato elements and Eisenstein symbols, as well as some new candidates for Sharifi's map in the group homology-to- K -theory direction.

Available at <https://xretep.github.io/pdfs/suslin.pdf>.

Presenting restricted Steinberg modules

Submitted. Defines “restricted” versions of Steinberg modules for general linear groups coming from local restrictions on cusps, and proves analogues of Borel–Serre's results relating their coinvariants with the geometry of partial compactifications of associated locally symmetric spaces. Gives generators and presentations for particularly nice cases of these restricted Steinberg modules.

Available at <https://xretep.github.io/pdfs/restricted.pdf>.

Eisenstein class of a torus bundle and log-rigid classes for $\mathrm{SL}_n(\mathbb{Z})$

Joint with Martí Roset Julià, submitted. Uses purely topological methods (part of the “topological polylogarithm” class, to be precise) to produce a $(n - 1)$ -cocycle for $\mathrm{SL}_n(\mathbb{Z})$ valued in differential forms

on Drinfeld’s p -adic upper half-space. Conjectures a relationship between evaluations of these classes and logarithms of Gross–Stark units, and proves this in some cases using the work of Dasgupta–Kakde. Available at <https://xretep.github.io/pdfs/RX1.pdf>.

Rigid analytic Eisenstein cocycles for $\mathrm{SL}_n(\mathbb{Z}[1/p])$

Joint with Martí Roset Julià, under revision. Uses symbols in a cohomological complex for a smooth torus to produce explicit $(n - 1)$ -cocycles for $\mathrm{SL}_n(\mathbb{Z}[1/p])$ valued in invertible analytic functions on Drinfeld’s p -adic upper half-space, generalizing existing results on rigid analytic cocycles from the case $n = 2$, in the “Eisenstein” setting. Relates the evaluations to periods of Eisenstein series. Proves that the special moduli of these cocycles yield Brumer–Stark units for totally real fields in which p is inert, using the integral Gross–Stark conjecture; provides evidence the general case relates to boundary degenerations of these same geometric objects.

Eisenstein cocycles for Drinfeld modular varieties.

Joint with Romyar Sharifi, in preparation. Constructs cocycles valued in products of the analogues of Siegel units on Drinfeld modular varieties, in motivic cohomology. Proves Hecke equivariance properties for these cocycles; in particular, in certain dimensions, constructing proving the Eisenstein property of Sharifi’s map ϖ_N relating the Eisenstein geometry of $\mathrm{GL}_n(A)$ and the motivic cohomology of a rank- $(n - 1)$ Drinfeld modular variety over A , for A any function field.

Classical periods of Eisenstein series and Bernoulli polynomials in the equivariant cohomology of a torus

Published in *Research in Number Theory*. Uses explicit elements in cohomology complexes of 2-tori to obtain a conceptual proof of a classical *unsmoothed* formula of Siegel for the periods of classical weight-2 holomorphic Eisenstein series. Available at <https://xretep.github.io/pdfs/classical.pdf>.

RESEARCH OUTREACH

Workshop on rigid cocycles

Oberwolfach, March 2027

Upcoming Oberwolfach seminar, whose material in part covers my joint work with Julià on rigid Eisenstein cocycles and their role in Hilbert's 12th problem for totally real fields. (Subject of the articles "Eisenstein class of a torus bundle..." and "Rigid analytic Eisenstein cocycles...").

Mixed motives and Steinberg modules

Eisenstein in ICTS Bengaluru, January 2027

Upcoming talk on my work, in relation to the role of Steinberg modules in recent developments in Goncharov's program and adjacent phenomena.

Combinatorial Eisenstein cocycles

2024 Bielefeld Workshop

Talk on the construction of explicit Eisenstein cocycles in the toric/elliptic cases (preprints mentioned above).

Higher Kato elements and equivariant complexes

2022 Quebec-Maine Number Theory

Talk on a part of my thesis work in a concrete setting.

Deformations of formal groups

2018 Montreal-Toronto Number Theory

Expository talk on the Gross-Hopkins construction of deformation spaces of formal groups.

Various seminar talks

Columbia, Caltech, UCSB, UCLA, etc.

Talks on various aspects of my research, over the span of a few years.

AWARDS AND SCHOLARSHIPS

McGill Graduate Excellence Award 2017 - 2023

Kenneth Eade Fellow 2018 - 2019

McGill Chair Graduate Award 2021 - 2022

TEACHING AND MENTORING

Lectureships

McGill University:

- MATH 141 Calculus II, Summer 2018 and Fall 2022

University of California, Los Angeles:

- MATH 131a Analysis, Fall 2023, Winter/Spring/Fall 2024, Winter/Fall 2025
- MATH 184 Enumerative Combinatorics, Spring 2024
- MATH 61 Introduction to Discrete Structures, Winter 2025
- MATH 111 Introduction to Number Theory, Spring 2025.

Directed Reading Program

Organized a mentorship program pairing McGill undergraduates interested in mathematics research with graduate students in corresponding fields. Solicited and paired graduate student volunteers with interested undergraduates, organized events, reviewed student expository write-ups, and participated as a mentor (in the first two years). The program has now expanded to be well-funded, with stipends for the graduate mentors, funds for resources for the students, and writing/presentation workshops.

UCLA–CSU Summer Bridge

August 2025. Prepared and taught an intensive one-week course (seven hours per day, with breaks) on representation theory of finite groups for a small group of students preparing to apply for doctorates in mathematics, coming from California State system universities which traditionally do not have strong programs for mathematical research. Wrote syllabus, assigned exercises, and gave personalized attention to each student throughout the long teaching sessions.